

TOPOLOGICAL (QUANTUM) FIELD THEORIES

IN QUANTUM COMPUTING

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MATH $\otimes$ QUANTUM

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OVERVIEW

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What is a qubit?

Spin systems, photons, superconducting circuits, trapped ions, . . .

But as theorists, we have a pretty clear answer.

A qubit consists of **vectors** of the form

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

If we consider all superpositions, we get a “space” of vectors we call $\mathbb{C}^2$.

This is the **Hilbert space** associated to a qubit q_0 .

The associated Hilbert space is the **tensor product** $\mathbb{C}^2 \otimes \mathbb{C}^2$.

$$q_0 : \mathbb{C}^2 \text{ —————}$$

$$q_1 : \mathbb{C}^2 \text{ —————}$$

More generally, say we have a quantum system consisting of n qubits.

Well, our picture now looks like

$$\begin{array}{l} q_0 : \mathbb{C}^2 \text{ ---} \\ q_1 : \mathbb{C}^2 \text{ ---} \\ \vdots \\ q_n : \mathbb{C}^2 \text{ ---} \end{array}$$

So, the Hilbert space associated to n qubits should be

$$\mathbb{C}^2 \underbrace{\otimes \dots \otimes}_{n} \mathbb{C}^2.$$

As we will see, this game of assigning Hilbert spaces to quantum systems based on tensor products is exactly what a topological quantum field theory is.

In the most generality, a Hilbert space is defined as follows.

Definition

A Hilbert space $\mathcal{H}$ is a complex inner product space that is complete with respect to the induced metric.

In **finite** dimensions, our Hilbert space is always $\mathcal{H} \simeq \mathbb{C}^n$. All this means is we care about vectors

$$|\psi\rangle = \alpha_1 |0\rangle + \alpha_2 |1\rangle + \cdots + \alpha_n |n-1\rangle$$

$$|\varphi\rangle = \beta_1 |0\rangle + \beta_2 |2\rangle + \cdots + \beta_n |n-1\rangle$$

... and being able to take inner products

$$\langle \psi | \varphi \rangle = \sum_{i=1}^n \alpha_i^* \beta_i.$$

In **infinite** dimensions, we care about the same sort of vectors, except now there are infinitely many terms! This means we have to be careful about when things converge.

An example of an infinite dimensional Hilbert space that crops up all the time in physics is L^2 space.

This is our space of all wave functions $\Psi(x, t)$.

For quantum computation (and for topological field theories), we “often” only care about the finite dimensional case.

Hilbert spaces also facilitate something like “time-reversal.”

Given an operator

$$M : \mathcal{H} \rightarrow \mathcal{K}$$

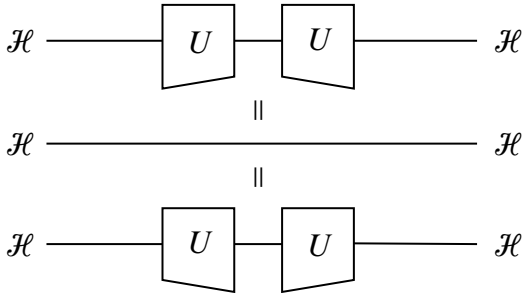
there is a **dagger** (or adjoint) operator

$$M^\dagger : \mathcal{K} \rightarrow \mathcal{H}.$$

A **quantum circuit** is an operator $U : \mathcal{H} \rightarrow \mathcal{H}$ such that

$$U^\dagger U = \mathbb{1} = U U^\dagger$$

Pictorially, this means



“Performing a circuit and then undoing it does nothing.”

To reason about quantum information processes mathematically, we tend to restrict ourselves using some **axioms**.

Let us consider an arbitrary quantum system Q .

The **state space** axiom says we model the quantum system Q by assigning to it a Hilbert space $\mathcal{H}_Q$.

For example, when Q is a collection of n qubits, we saw that

$$\mathcal{H}_Q \simeq \mathbb{C}^2 \underbrace{\otimes \cdots \otimes}_{n} \mathbb{C}^2.$$

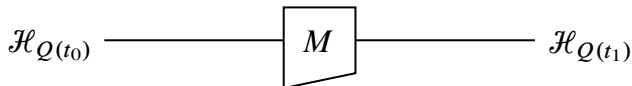
Now, say our system $Q(t)$ is time-dependent.

How would we represent time evolution

$$Q(t_0) \longrightarrow Q(t_1)?$$

Well by the state space axiom, we have a Hilbert space $\mathcal{H}_{Q(t_0)}$ and a Hilbert space $\mathcal{H}_{Q(t_1)}$.

So, the **evolution** axiom says we model evolution using operators



Moreover, if the evolution is closed/reversible, we ask that our operator be a quantum circuit. This property is called **unitarity**.

Suppose now that we want to combine or **correlate** two systems.

How should we go about modeling this?

Let A be Alice's system and B be Bob's. To consider them as a single quantum system, we want to consider the **joint** system

$$A \amalg B.$$

The **multiple systems** axiom says that the state space associated to the joint system of Alice and Bob is just the tensor product of their state spaces:

$$\mathcal{H}_{A \sqcup B} \simeq \mathcal{H}_A \otimes \mathcal{H}_B.$$

$$\mathcal{H}_{A \amalg B} \quad \text{---} \quad = \quad \begin{array}{c} \mathcal{H}_A \text{---} \\ \mathcal{H}_B \text{---} \end{array}$$

$$\text{Quant} \rightarrow \mathcal{H}\text{ilb.}$$

What is a **setting**?

Well, we should have

- (i) things “objects.”
- (ii) processes “morphisms.”
- (iii) successive processes “composition.”

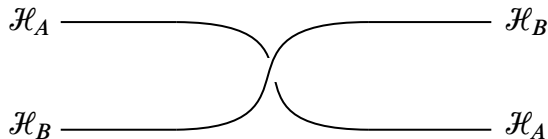
Formally, we usually call settings **categories**.

Then, we call a change of setting a **functor**.

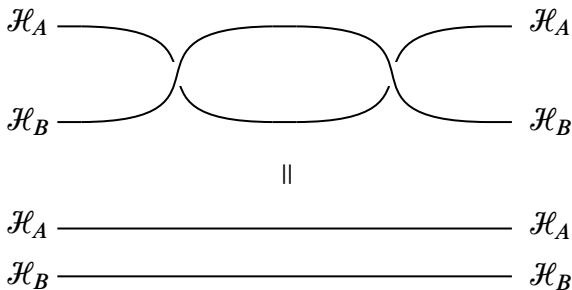
Sometimes we can equip our setting with a way to join objects and processes:

- (i) in $\mathcal{H}\text{ilb}$ we have $\otimes$.
- (ii) in $\mathcal{Q}\text{uant}$ we have $\amalg$.

In fact, we can sometimes swap joined objects:



In our two settings, swapping twice does nothing (“symmetric”):



1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

A **functorial quantum field theory** is a symmetric monoidal functor

$$\mathcal{Z} : \mathcal{Q}uant \rightarrow \mathcal{H}ilb.$$

That is, $\mathcal{Z}$ is a change in setting that respects the joining operation.

We can choose all sorts of models for $\mathcal{Q}$ uant.

In typical **topological field theory** (in dimension $d + 1$), we let

$$\mathcal{Q}uant = \mathcal{B}ord_{d+1}.$$

The monoidal category $\mathcal{B}ord_{d+1}$ has

- (i) objects that are d -dimensional physical spaces.
- (ii) processes given by $(d + 1)$ -dimensional “spacetimes.”
- (iii) composition is gluing.
- (iv) joining is just using $\amalg$.

Let us consider some examples of these processes in dimension 2.

Here are some classic spacetimes between circles:

- pair of pants (2 to 1)



- cup (o to i)



- swap (2 to 2)



So, a topological quantum field theory in dimension $d + 1$ is a symmetric monoidal functor

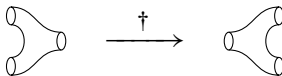
$$Z : \mathcal{B}ord_{d+1} \rightarrow \mathcal{H}ilb.$$

Notice that we have not said anything about **closed evolution**.

To encode unitarity into a topological field theory, we can add a **dagger** to give us time reversal.

For $\mathcal{H}\text{ilb}$, we already saw the dagger!

In dimension 2, this looks like



Then a **unitary** topological field theory is a symmetric monoidal $\dagger$ -functor

$$\mathcal{Z} : \mathcal{B}ord_{d+1} \rightarrow \mathcal{H}ilb$$

This is a change in setting that respects the joining operation and the $\dagger$ operation.

Since $\mathcal{Z}$ respects the $\dagger$, it takes reversible evolutions to quantum circuits. As you may expect, these are often the topological field theories we care about in quantum information science.

So . . . , where can we go from here?

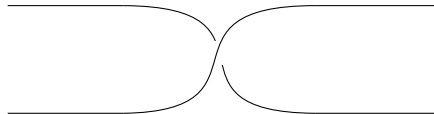
Well, we saw that the pictures we drew of our modeling processes gave us the notation for quantum circuits.

We can study all sorts of settings with various graphical calculi.

These diagrams give an interesting unifying perspective on quantum theory.

For example, **Feynman diagrams** in particle physics are exactly the graphical calculus of a certain monoidal category (see Baez-Lauda for more on this).

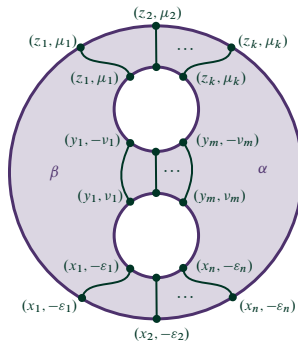
Recall that we often have a way to swap systems:



Sometimes this **braiding** operation is not symmetric, so doing it twice does not return us to our original state.

It turns out that in the setting of **anyons**, we can use these braidings and “fusion rules” to encode our quantum gates (see Burton, Preskill, Kawagoe for more on this).

Here is one defect pair of pants in $\mathcal{Def}_2$:



It turns out,¹ the resultant (dimension 2) defect field theories

$$\mathcal{Z} : \mathcal{Def}_2 \rightarrow \mathcal{Hilb}$$

give rise to gadgets called pivotal 2-categories. These are settings with “processes between processes.”

¹This construction is due to Davydov, Kong, and Runkel (link).

I learned it from some notes by Nils Carqueville (link).

Further reading:

- (BL) John Baez and Aaron Lauda's chapter *A Prehistory of n -Categorical Physics* (link).
- (B) Simon Burton's note *A Short Guide to Anyons and Modular Functors* (link).
- (K) Kyle Kawagoe's lecture notes *Fusion Categories and Condensed Matter* (link).
- (P) John Preskill's chapter *Topological Quantum Computation* (link).

